

Example 12: Express the statement "Everyone has exactly one best friend" as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Soln: $B(x, y)$: y is the best friend of x .
 $\forall x \forall y \forall z (B(x, y) \wedge (y \neq z) \Rightarrow \neg(B(x, z)))$ $\rightarrow$ My solution

Book solution

The statement "Everyone has exactly one best friend" can be expressed as "For every person x , person x has exactly one best friend." Introducing the universal quantifier, we see that this statement is the same as " $\forall x$ (person x has exactly one best friend)," where the domain consists of all people.

To say that x has exactly one best friend means that there is a person y who is the best friend of x , and furthermore, that for every person z , if person z is not person y , then z is not the best friend of x . When we introduce the predicate $B(x, y)$ to be the statement " y is the best friend of x ," the statement that x has exactly one best friend can be represented as $\exists y (B(x, y) \wedge \forall z ((z \neq y) \Rightarrow \neg B(x, z)))$.

Consequently, our original statement can be expressed as

$$\forall x \exists y (B(x, y) \wedge \forall z ((z \neq y) \Rightarrow \neg B(x, z)))$$

Example 13: Use quantifiers to express the statement "There is a woman who has taken a flight on every airline in the world."

Soln: $W(x)$: x is a woman. $AF(x, y)$: y is a flight of airline x .
 $F(x, y)$: Person x has taken flight y .
 $\exists x \forall y \exists z (W(x) \wedge AF(y, z) \wedge F(x, z))$, where x is taken from the domain of people, y is taken from the domain of airlines and z is taken from the domain of flights.